

Curve Fitting by Least Squares Method, Correlation and Regression

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Curve Fitting (by Least Squares Method)

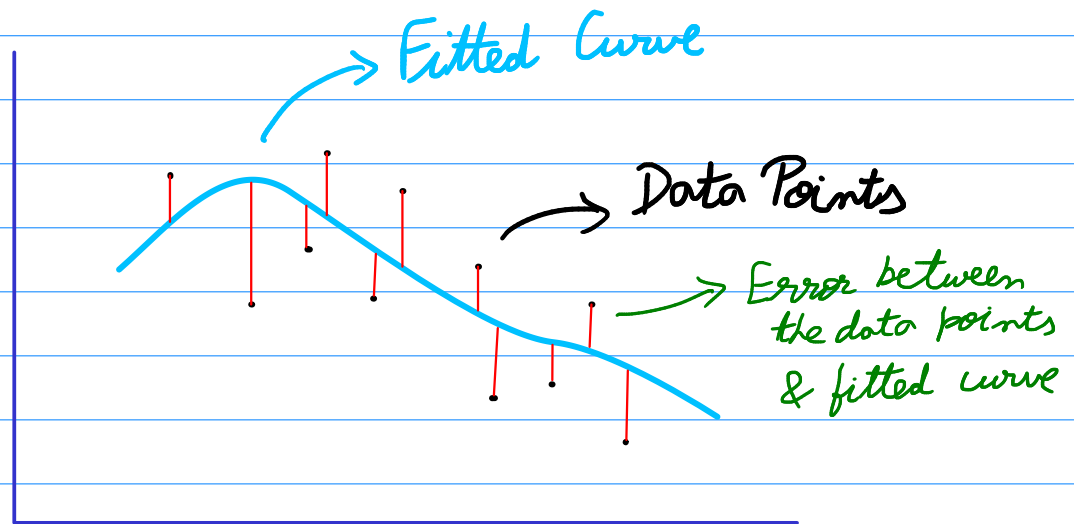
Given data points $\{(x_i, y_i)\}_{i=1}^n$ the objective of curve fitting is

To find a curve that best 'fits' the data.

What we mean is that we do NOT want a curve that "passes" through the given data points (recall this is what we did in interpolation)

Instead we want a curve that MAY or MAY NOT pass through the data points, but the error between the data points and the curve is MINIMUM.

AIM
We want to find the curve such that the errors are minimised



Why curve fitting? Why not find a curve passing through all points?

In Statistics we deal with data that has been collected via experiments, surveys etc - sampling. Now an inherent problem with this is that there may be errors in the data - errors of measurement, or errors in filling forms for survey etc.

So, we DON'T want to blindly trust the data.

But at the time our data is not useless. In general, even if there are errors in the data those errors will be random - some will be positive errors and some will be negative errors - and for a truly random sample, these errors will average out to be close to 0.

So, we can still use this data to capture the 'general trend' of the data i.e. do the data points lie around a straight line, or around a parabola or around an exponential curve etc.

This is the aim of curve fitting.

How to fit a curve?

The basic idea is simple. Choose a curve to fit $y = f(x)$

Example - a straight line $y = ax + b$

a parabola $y = ax^2 + bx + c$

an exponential curve $y = ae^{bx}$ or $y = ax^b$

and many others

How do statisticians practically choose which curve to use?

They may study the data by plotting it as a scatter plot and make a guess, or they may use other measures of correlation and other statistical concepts

Now the equation of the curve has constants $a, b, c, \dots$ which need to be determined. For this we first compute the 'errors' by subtracting the recorded value for y and the value coming from our chosen function.

$$\text{Error at } x_i = |y_i - f(x_i)|$$

A more useful measure is to take the squares of the errors

$$\text{Squared Error at } x_i = \left(y_i - f(x_i) \right)^2$$

Now we sum up these errors and try to minimise them

This is why the method is called LEAST SQUARES METHOD

We are trying to find the curve such that the SQUARES of the errors are LEAST.

Fit a straight line to the given data

Assume the line to be fitted is $y = ax + b$

Now for each point (x_i, y_i) in the data we have $(x_i, y(x_i))$ is the corresponding point on the line where $y(x_i) = ax_i + b$

Thus, the total error is

$$S(a, b) = \sum_{i=1}^n (y(x_i) - y_i)^2 = \sum_{i=1}^n (ax_i + b - y_i)^2$$

Note here S is a function of a & b as these are the unknowns.

All else is given to us.

We now simply want to find a & b so that $S(a, b)$ is minimum. From the 'Theory of Functions of Several Real Variables', the condition for this is

$$\frac{\partial S}{\partial a} = 0 \quad \& \quad \frac{\partial S}{\partial b} = 0$$

i.e. we have the equations

$$\sum_{i=1}^n 2x_i(ax_i + b - y_i) = 0$$

$$\& \sum_{i=1}^n 2(ax_i + b - y_i) = 0$$

i.e.

$$\boxed{\begin{aligned} a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i - \sum_{i=1}^n x_i y_i &= 0 \\ \& \quad a \sum_{i=1}^n x_i + nb - \sum_{i=1}^n y_i &= 0 \end{aligned}}$$

This gives us a system of two linear equations in a & b which can be solved to find a & b .

Fit other special type of curves by converting to 'linear' or straight line form

Suppose we have to fit the curve of type $y = ae^{bx}$ to the data.

Instead of trying to find a & b by trying to minimise

$$S(a, b) = \sum_{i=1}^n (y(x_i) - y_i)^2 = \sum_{i=1}^n (ae^{bx_i} - y_i)^2$$

which would result in complicated, hard to solve equations involving e , we can convert $y = ae^{bx}$ as follows
Take natural log on both sides

$$\ln y = \ln(ae^{bx}) = (\ln a) + bx$$

which is of the form

$$Y = A + BX$$

where $Y = \ln y$, $X = x$, $A = \ln a$, $B = b$

Now instead of trying to 'fit' $y = ae^{bx}$ to $\{(x_i, y_i)\}_{i=1}^n$; we can fit $Y = A + BX$ to the data $\{(x_i, \ln y_i)\}_{i=1}^n$.

Then $a = e^A$, $b = B$ will give the required values of $y = ae^{bx}$.

Similarly we may change other curves like

$$y = ax^b, \quad y = bx^a \quad (\text{Take log})$$

$$y = a + \frac{b}{x} \quad (\text{Take lcm})$$

Fit a polynomial to the given data

Assume the curve to fit is $y = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$

Here we have to minimise the function

$$\begin{aligned} S(a_0, a_1, \dots, a_n) &= \sum_{i=1}^n (y(x_i) - y_i)^2 \\ &= \sum_{i=1}^n (a_0 + a_1x_i + a_2x_i^2 + \dots + a_nx_i^n - y_i)^2 \end{aligned}$$

Then, taking $\frac{\partial S}{\partial a_0} = \frac{\partial S}{\partial a_1} = \dots = \frac{\partial S}{\partial a_n} = 0$ we get

the $(n+1)$ equations

$$\begin{cases} na_0 + a_1 \sum_{i=1}^n x_i + a_2 \sum_{i=1}^n x_i^2 + \dots + a_n \sum_{i=1}^n x_i^n - \sum_{i=1}^n y_i = 0 \\ a_0 \sum_{i=1}^n x_i + a_1 \sum_{i=1}^n x_i^2 + a_2 \sum_{i=1}^n x_i^3 + \dots + a_n \sum_{i=1}^n x_i^{n+1} - \sum_{i=1}^n x_i y_i = 0 \\ a_0 \sum_{i=1}^n x_i^2 + a_1 \sum_{i=1}^n x_i^3 + a_2 \sum_{i=1}^n x_i^4 + \dots + a_n \sum_{i=1}^n x_i^{n+2} - \sum_{i=1}^n x_i^2 y_i = 0 \\ \vdots \\ a_0 \sum_{i=1}^n x_i^n + a_1 \sum_{i=1}^n x_i^{n+1} + a_2 \sum_{i=1}^n x_i^{n+2} + \dots + a_n \sum_{i=1}^n x_i^{2n} - \sum_{i=1}^n x_i^n y_i = 0 \end{cases}$$

Solving this linear system of equations by Gauss Elimination or Gauss Jordan method, we can find the values of $a_0, a_1, \dots, a_n$.

SS Sastry

- 4.1 Explain the method of least squares to fit a straight line of the form $Y = a_0 + a_1x$ to the data (x_i, y_i) :

x	1	2	3	4	5	6
y	2.4	3.1	3.5	4.2	5.0	6.0

- 4.2 Find the values of a_0 and a_1 so that $Y = a_0 + a_1x$ fits the data given in the table:

x	0	1	2	3	4
y	1.0	2.9	4.8	6.7	8.6

- 4.5 Find the values of a, b, c so that $Y = a + bx + cx^2$ is the best fit to the data:

x	0	1	2	3	4
y	1	0	3	10	21

- 4.6 Fit a least squares parabola $Y = a + bx + cx^2$ to the data:

x	0	1	2	3	4	5	6
y	71	89	67	43	31	18	9

- 4.7 Determine the normal equations if the cubic polynomial $Y = a_0 + a_1x + a_2x^2 + a_3x^3$ is fitted to the data (x_i, y_i) , $i = 1, 2, \dots, m$.

- 4.8 Determine the constants a and b , by the method of least squares, such that the curve $y = ae^{bx}$ fits the data:

x	2	4	6	8	10
y	4.077	11.084	30.128	81.897	222.62

- 4.9 Fit a function of the form $y = ax^b$ for the following data:

x	61	26	7	2.6
y	350	400	500	600

- 4.10 Using the method of least squares, fit a curve of the form $Y = \frac{a}{x} + bx$ to the following data (x, y) :

$(1, 5.43), (2, 6.28), (4, 10.32), (6, 14.86), (8, 19.51)$.

Correlation

In real world scenarios, often we have no idea what are the variables that affect the phenomenon we are studying.

For example - suppose we want to study what affects the bird population density in an area - we do not know what affects the bird population to start with - so we make guesses as to what factors might be important.

Suppose we guess, that tree cover affects the bird population, then the next step is to go in the field and collect data of tree cover and the bird population in different areas.

After this step, we get a table of data. This data cannot tell you if our guess was correct and whether the tree cover density actually affects the bird population.

How do we start to confirm our guess?

One way is to observe if the values of tree cover (x) and bird population (y) are somehow dependant on each other?

Does y increase or decrease as x increases?

This is called 'correlation' between x and y .

Thus, correlation is a way to measure when two variables are 'possibly related' to each other - by measuring to what extent does the value of y seem to change depending on the change in x .

But of course just because there seems to be a pattern in the way the values of two variables change - does NOT mean that the variables must be related or that one must be the cause of the other.

For example - suppose x denote the number of people suffering from obesity and diabetes in a city while y represents the number of health centres and hospitals in the city.

Then, it may happen that cities with higher number of health facilities report a higher caseload of people suffering from obesity and diabetes, because a greater number of people are correctly diagnosed as opposed to cities where due to a lack of health facilities, many patients of diabetes may go undiagnosed.

So if we observe the patterns we will see that the values of x seem to increase with the values of y .

But this is NOT a true relation, as certainly a person's risk of developing diabetes does NOT depend on the number of hospitals in the city where they live.

So, while correlation tells us when there may be a relation between two variables, it is NOT the final test.

Just because two variables have high correlation, does NOT mean that one variable is the cause of the other, which brings us to the statistical maxim

" CORRELATION IS NOT CAUSATION "

which just sums up the above discussion.

How to measure correlation?

There are different measures of correlation - each one is designed to measure a certain aspect of relationship between two variables.

We will be concerned with two main types

- Pearson's correlation coefficient
- Spearman's rank correlation coefficient

Pearson's correlation coefficient

Pearson's correlation coefficient measures how strongly the two variables are related by a linear relationship and in which direction.

By a linear relationship between two variables x and y , we mean a relationship of the form $y=ax+b$, where a and b are some constants.

The values of Pearson's correlation coefficient lies between -1 and 1

A value of

- 1 means x and y lie perfectly on a straight line of positive slope
i.e. y increases as x increases and the relationship is given by $y=ax+b$, where $a>0$
- -1 means x and y lie perfectly on a straight line of negative slope
i.e. y decreases as x increases and the relationship is given by $y=ax+b$, where $a<0$
- 0 means there is no linear relationship between x and y.
This may happen if for instance y depends periodically on x for example, if $y=\sin(x)$.

A value between

- 0 and 1 means x and y have some sort of linear relationship such that y increases as x increases. The closer the value is to 1, more 'linear' is the relationship. Example, $y=e^x$, $y=x^2$ etc. are all relations that are NOT perfectly linear but have some positive linear component to it.
- -1 and 0 means x and y have some sort of linear relationship such that y decreases as x increases. The closer the value is to -1, more 'linear' is the relationship. Example, $y=e^{-x}$, $y=-x^3$ etc. are all relations that are NOT perfectly linear but have some negative linear component to it.

Formula for Pearson's correlation coefficient

Given the data $\{(x_i, y_i)\}_{i=1}^n$ the Pearson's correlation coefficient is defined as,

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

$$\begin{aligned} &\text{Schwartz Ineq} \\ &|\langle u, v \rangle|^2 \leq \|u\|^2 \|v\|^2 \\ &\Rightarrow -1 \leq r_{xy} \leq 1 \end{aligned}$$

where $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ & $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$

are the means of x_i & y_i .

NOTE that $r_{xy} = r_{yx}$

Spearman's rank correlation coefficient

Spearman's rank correlation coefficient measures how strongly the two variables are related by a monotonic relationship and in which direction.

By a monotonic relationship between two variables x and y , we mean an increasing or decreasing relationship of any form, example an exponential relationship $y = e^x$, $y = x^7$, $x = y^{1/3}$, $y = e^{-x}$, $y = -x^{15}$ etc. or a logarithmic relationship $y = \ln x$ etc.

The values of Spearman's rank correlation coefficient lies between -1 and 1

A value of

- 1 means x and y lie perfectly on an increasing monotonic function, i.e. y always increases whenever x increases
- -1 means x and y lie perfectly on a decreasing monotonic function, i.e. y always decreases whenever x increases
- 0 means there is no monotonic relationship between x and y .
This may happen if for instance y depends periodically on x for example, if $y = \sin(x)$.

A value between

- 0 and 1 means x and y have some sort of increasing monotonic relationship i.e. y generally increases as x increases. The closer the value is to 1, more 'monotonic increasing' is the relationship.
- -1 and 0 means x and y have some sort of decreasing monotonic relationship i.e. y generally decreases as x increases. The closer the value is to -1, more 'monotonic decreasing' is the relationship.

How to calculate Spearman's rank correlation coefficient

Given the data $\{(x_i, y_i)\}_{i=1}^n$, we rank the x_i 's and y_i 's from 1 to n in increasing order

Egⁿ Suppose we are given $(-3, 5)$, $(8, 63)$, $(7, -32)$, $(0, 0)$ then we rank the data as

RANKS of x_i 's	1	4	3	2
x_i	-3	8	7	0
y_i	5	63	-32	0
RANKS of y_i 's	3	4	1	2

Now Spearman's rank correlation coefficient of x & y is defined as the Pearson's correlation coefficient of (RANKS of x) & (RANKS of y).

Egⁿ Suppose we are given $(-3, 5), (8, 63), (7, -32), (0, 0)$

then $\bar{x} = \frac{1}{4}(-3+8+7+0) = 3$

$\bar{y} = \frac{1}{4}(5+63-32+0) = 9$

x_i	y_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$y_i - \bar{y}$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
-3	5	-6	36	-4	16	24
8	63	5	25	54	2916	270
7	-32	4	16	-41	1681	-164
0	0	-3	9	-9	81	27
			<u>86</u>		<u>4694</u>	<u>157</u>

Then Pearson's correlation coefficient is

$$= \frac{157}{\sqrt{86} \sqrt{4694}} \approx 0.247$$

Now for Spearman's rank correlation coefficient we have

$X \leftarrow$	RANKS of x_i 's	1	4	3	2
	x_i	-3	8	7	0
	y_i	5	63	-32	0
$Y \leftarrow$	RANKS of y_i 's	3	4	1	2

Let X & Y denote the ranks of x_i 's & ranks of y_i 's respectively

$$\bar{X} = \frac{1}{4}(1+4+3+2) = 2.5$$

$$\bar{Y} = \frac{1}{4}(3+4+1+2) = 2.5$$

x_i	y_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$y_i - \bar{y}$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
1	3	-1.5	2.25	0.5	0.25	-0.75
4	4	1.5	2.25	1.5	2.25	2.25
3	1	0.5	0.25	-1.5	2.25	-0.75
2	2	-0.5	0.25	-0.5	0.25	0.25
			5			5
						1

Then the Spearman's rank correlation coefficient is

$$= \frac{1}{\sqrt{5} \sqrt{5}} = \frac{1}{5} = 0.2$$

Thus the Pearson's correlation coefficient comes out to be 0.247, which means there is a positive linear relationship between x & y but it is weak as the value is closer to 0 than 1

Similarly the Spearman's rank correlation coefficient is 0.2, which means there is a monotonic increasing relationship between x & y but it is weak as the value is closer to 0 than 1.

Note

If the data contains repetitions, then the ranks to be awarded is the fractional rank i.e. the average of the ranks awarded to the repeating value if repetition is ignored

For example - if $x = 0, 1, 1, 2, 3, 4, 5, 5, 5, 6$

Then if we rank x without counting repetition the ranks will be
1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Now as $x=1$ occurs twice and has ranks 2 and 3 - we give it the rank $(2+3)/2=2.5$

Similarly as $x=5$ occurs thrice and has the ranks 7,8,9 - we give it the rank $(7+8+9)/3=8$

So the ranks become 1, 2.5, 2.5, 4, 5, 6, 8, 8, 8, 10

Regression

Regression is the study of obtaining an actual relationship between two variables x and y . Unlike correlation, which is used to measure the direction (increasing or decreasing) and nature of relationship (if any) between two variables, regression actually computes the relation.

The simplest type of regression is simple linear regression, in which we are dealing with only two variables and want to find the equation of line that expresses y in terms of x or the other way around.

The equation of line that expresses y in terms of x is called the line of regression of y on x

The equation of line that expresses x in terms of y is called the line of regression of x on y

There are many ways to find the line of regressions - the most commonly used method is the method of curve fitting.

For the line of regression of y on x , we aim to fit a line $y=ax+b$ to the data by minimising the squares of the errors in y .

For the line of regression of x on y , we aim to fit a line $x=ay+b$ to the data by minimising the squares of the errors in x .

We will be only concerned with simple linear regression between two variables x and y .

Equation of Line of Regression of x on y Taking x as dependent variable and y as independent variable

$$x - \bar{x} = b_{yx}(y - \bar{y})$$

where

$$\bar{x} = \frac{1}{n} \sum x_i$$

Mean of x_i 's

$$\bar{y} = \frac{1}{n} \sum y_i$$

Mean of y_i 's

b_{yx} is the regression coefficient of x on y and is given by

$$b_{yx} = r \left(\frac{\sqrt{\sum (x_i - \bar{x})^2}}{\sqrt{\sum (y_i - \bar{y})^2}} \right)$$

Pearson's Correlation Coefficient of x and y

Equation of Line of Regression of y on x Taking y as dependent variable and x as independent variable

$$y - \bar{y} = b_{xy} (x - \bar{x})$$

where

$$\bar{x} = \frac{1}{n} \sum x_i$$

Mean of x_i 's

$$\bar{y} = \frac{1}{n} \sum y_i$$

Mean of y_i 's

b_{yx} is the regression coefficient of y on x and is given by

Note that if $|r|=1$ then both slopes are reciprocals of each other - which means fitting a line for x on y and for y on x gives the same line.

$$b_{xy} = r \left(\frac{\sqrt{\sum (y_i - \bar{y})^2}}{\sqrt{\sum (x_i - \bar{x})^2}} \right)$$

Pearson's Correlation Coefficient of x and y

However, if $|r| < 1$ then curve fitting a straight line gives different lines for x on y and y on x. But the slopes are still related by r

Note that $r = \sqrt{b_{xy} b_{yx}}$

Also as $-1 \leq r \leq 1$ we have $0 \leq r^2 \leq 1$ thus $0 \leq b_{xy} b_{yx} \leq 1$